

A Denoising and Drift-control Approach for UAV Trajectory Tracking

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Abstract—As part of dead reckoning, inertial navigation has been developed for many years in locating the vehicle without GPS. We employ the technology on a quadrotor platform to track the trajectory of the vehicle. In this paper, a new method was designed to make the tracking result more precisely. To achieve this, we firstly smooth the sensor data by minus offset dynamically and adopt Kalman Filtering algorithm. Then we detect velocity and correct the wrong value in time to decrease drift in displacement. This is an important step in locating the vehicle simply by odometry.

I. INTRODUCTION

The Unmanned Aircraft Vehicle (UAV) has become more and more popular in recent years. Because its low-cost and flexible property, this aircraft is widely used not only in military areas but also in civil areas such as weather reconnaissance, agriculture environmental monitoring[1]. Many exteroceptive sensors such as radar, ultrasonic and RGB-D camera, and interoceptive sensors such as accelerometer, gyroscope and magnetometer have been used in UAV to manage its flight and perceive the surroundings. The key technics involved in UAV includes navigation, simultaneously localization and mapping (SLAM), autonomous flight, etc.

Quadrotor is one of the most common type of UAV. A quad-rotor employs four fixed propellers to achieve its movement. It has four inputs but has six degree of freedom (DOF), which makes the aircraft an inherently under-actuated system. Lots of research on control method of the UAV has been developed. The control method such as basic PID control method[2], method based on back-stepping algorithm[3] and output feedback control using neural networks[4] have been investigated and tested in various UAVs. Recently the control method designed for controlling multi-agent have gained much interest, different kinds of algorithms toward this topic are presented[5][6][7].

UAV performs well in some complicate circumstance that ground mobile robot may encounter some problems. Some research topic, for example, 3D SLAM and version-based navigation[8], are developed using the camera equipped on this vehicle to expand the sensing angle and coverage comparing with the mobile robot. Sensing data such as images from UAV can be used not only for self-control and navigation, but also for cooperative sharing to other vehicles (e.g. the ground mobile robot) to help them understand their

surroundings. In [9], they combine the UAV with the ground robot to make a working system. Data from the UAV is used to perceive the environmental information and lead the multi-mobile-robot to achieve some complex goal.

Nowadays some popular research platforms are developed, such as STARMAC of Stanford University[10], humming bird of Asc Tech Ltd[11], AR-Drone by Parrot Ltd[10], crazyfly of Bitcraze Ltd. These vehicles behave well in some complex conditions, which is extremely important because it can release us from some complicate trouble that is unrelated to our aim[12].

Much contribution have been made in making quadrotor an automatically flight. Navigation based on SLAM is one useful way to achieve this goal[13]. However, high-accuracy sensor, such as the laser sensor is not always useful for some quadrotor. Giant computing capacity is also needed, which restrain extensive use of this method. In [14], a fully autonomous indoor quadrotor is developed simply based on camera. For micro aircraft vehicle(MAV), they can not carry much load generally, therefore any device equips for this platform is worth considering. [15] provide a novel method for localization of quadrotor simply using inertial measurement unit(IMU). They make use of the dead reckoning algorithm and correct the error to control drift.

The IMU data provides the corner stone for UAV to adjust the posture and fly under-control. Especially for the MAV, such as crazyfly, it is unstable inherently than bigger ones and can be disturbed more by environment so they can often fly out of control. This paper provided a method of minimizing the error when we use data of IMU to draw the trajectory of the quadrotor. This works mainly in two ways, smoothing the data of sensors and detecting velocity and displacement in real-time when tracking the trajectory of the vehicle.

This paper is organized as follows, section 2 introduces the structure of the quadrotor, including its control principle and modeling. Section 3 presents the method that we use to suppress drift. Section 4 gives the experiment setting and result. Section 5 concludes our work and presents relate work can be done in the future.

II. QUADROTOR PLATFORM

A. Basic Structure

Quadrotor is one kind of UAV that has many advantages than other fixed-wing or vertical take-off and landing vehicles (VTOL). It has three, four, six or other numbers of propellers that provide basic propulsion. This vehicle achieves three-dimensional mobility through changing the propulsion. Fig.1 is the model of quadrotor, as the figure shows, the rotation direction of motor 1 and motor 3 is

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different with that of motor 2 and motor 4, this design is employed to eliminate the aerodynamic torque effect of the motor. The motor, can be coreless motor or brushless motor, is determined by the controller through some kind of control method.

B. Control Principle

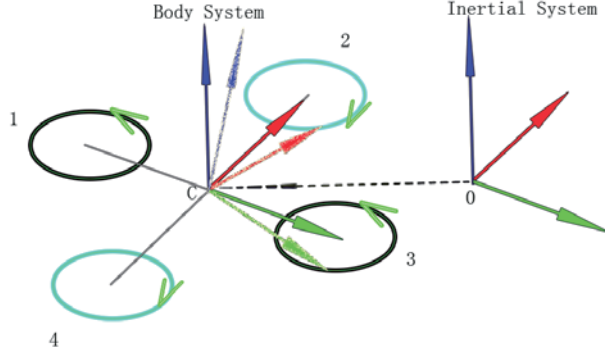


Fig. 1. Two kind of coordinate frame of the quadrotor

Sensors like IMU on board can provide the gesture and position information. This data is useful in controlling the quadrotor, although it can not be used directly. In this process, two kind of coordinate system is introduced. One is fixed coordinate system, called inertial frame, and the other is called body frame. As Fig.1 shows, the data acquired from the sensors is referring to the body system, however, the control method is based on the inertial system. Therefore we should do some transformation between these two system. Quaternion is one useful tool to transfer the data from one system to other[16]. It is denoted as follows:

$$Q = (q_0, q_1, q_2, q_3) = q_0 + q_1i + q_2j + q_3k \quad (1)$$

where $q_0 = \cos(\varphi/2)$ and φ mean the roller angle, the transform matrix is denoted as follows:

$$\begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & q_0^2 + q_2^2 - q_1^2 - q_3^2 & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & q_0^2 - q_2^2 + q_1^2 - q_3^2 \end{pmatrix} \quad (2)$$

The euler angle gives the angle of raw, pitch, and yaw angle of the vehicle referring to the fixed system, which could be used to adjust the motion of the quadrotor on the contrary.

C. Modeling

A model of quadrotor is established so as to make it easier to analyze the motion of this vehicle. Claim that $E_q = \{e_x, e_y, e_z\}$ be the coordinates of the inertial frame and the $B_q = \{b_x, b_y, b_z\}$ be the body system fixed on the quadrotor. Then the dynamic model of the quadrotor can be explicated as the following equations[17]:

$$\dot{\alpha} = v \quad (3)$$

$$\dot{v} = ge_z - TRe_z/m \quad (4)$$

$$\dot{R} = RS(\varpi) \quad (5)$$

$$I_f \dot{\varpi} = -\varpi \times I_f \varpi - G_a + \tau_a \quad (6)$$

$$I_r \dot{\omega}_i - Q_{i,i} \in \{1, 2, 3, 4\} \quad (7)$$

where m is the mass of the quadrotor and g is the gravitational acceleration. $v = (v_x, v_y, v_z)^T$ is the linear velocity of the body system. $\varpi = (\varpi_1, \varpi_2, \varpi_3)^T$ denotes the angular velocity of the airframe frame expressed on the body system. $S(\varpi)$ is the skew symmetric matrix that is denote by the following equations:

$$\varpi = \begin{pmatrix} 0 & -\varpi_3 & \varpi_2 \\ \varpi_3 & 0 & -\varpi_1 \\ -\varpi_2 & \varpi_1 & 0 \end{pmatrix} \quad (8)$$

and R represent the transform matrix of the two systems

$$S(\varpi) = \begin{pmatrix} C_\theta C_\psi & C_\psi S_\theta S_\phi - S_\psi C_\phi & C_\psi S_\theta C_\phi + S_\psi S_\phi \\ C_\theta S_\psi & S_\psi S_\theta S_\phi + C_\psi C_\phi & S_\psi S_\theta C_\phi - C_\psi S_\phi \\ -S_\theta & S_\phi C_\theta & C_\phi C_\theta \end{pmatrix} \quad (9)$$

I_f is the matrix of rotational inertia, G_a and τ_a donate the torque of the gyroscope and the fuselage respectively. $Q_i = k\omega^2$ is the resistance. From (3) to (7), the model of the quadrotor can be depicted as the following equations:

$$\ddot{\phi} = \dot{\theta} \dot{\psi} \left(\frac{I_y - I_z}{I_x} \right) - \frac{J_R}{I_x} \dot{\theta} + \frac{L}{I_x} U_2 \quad (10)$$

$$\ddot{\theta} = \dot{\phi} \dot{\psi} \left(\frac{I_z - I_x}{I_y} \right) + \frac{J_R}{I_y} \dot{\phi} + \frac{L}{I_y} U_3 \quad (11)$$

$$\ddot{\psi} = \dot{\phi} \dot{\theta} \left(\frac{I_x - I_y}{I_z} \right) + \frac{1}{I_z} U_4 \quad (12)$$

where ϕ θ ψ represent the angle of roll, pitch and yaw respectively. I_x, I_y, I_z is the rotational inertia of three axis respectively. J_R is the rotational inertia of the motor. L gives the distance between the center of gravity and motor. U_1, U_2, U_3 represent the input of three orientations:

$$\begin{cases} U_2 = k(-\omega_1^2 - \omega_2^2 + \omega_3^2 + \omega_4^2) \\ U_3 = k(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2) \\ U_4 = k(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{cases} \quad (13)$$

III. DATA OPTIMIZATION

Displacement of three axis can be combined to represent the location in the space, thus we can draw the trajectory of the vehicle using these data. In our system, sensors on board provide various data, such as the IMU, especially the barometer. As the Fig.2 shows, the raw data of three axis accelerators are not always change their value around the 0 axis. On the other hand, when the quadrotor flies for a period of time, the regression line of three axis shows random shift.

During the process of calculating the acceleration of Z axis. Barometer data is applied to fuse with accelerator data to correct the drift of IMU. This is useful because it reduces the drift and shows great property of tracking when calculating the displacement of Z axis. However, there is no other sensors equipped so the drift may be large on X and Y axis if we just rely on sensor data.

We divided the drift into two categories through analyzing acceleration data of three axis. One is initialization noise,

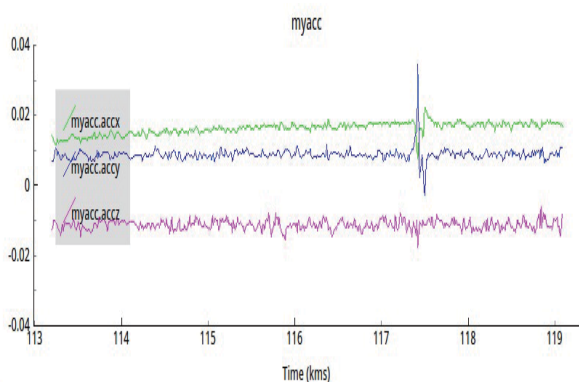


Fig. 2. Drift when system initialization

that introduces drift because different sensor may have offset inherently, thus we need to do some calibration before we use it. On the other hand, much noise will be introduced into the system and causing the drift of some axis when the quadrotor flying. Based on these analysis, we do our work in two aspects.

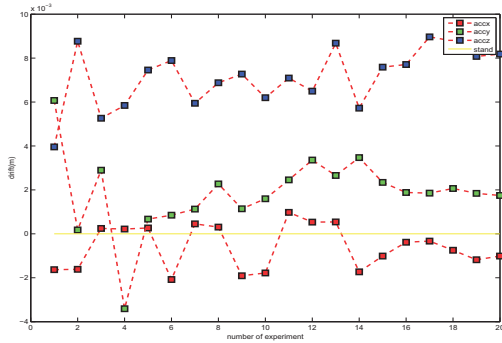


Fig. 3. Drift error in 20 experiment when the system initialization

We collected acceleration data of three axis to help ensure the property of the drift at the beginning. In Fig.3, data represent the drift error of three axis in the first 10 second when the system begins in 20 experiment. It shows that for certain IMU, the difference of drift error among different experiments can be small, this means it is possible to minus the offset value so as to correct the IMU data. Setting a initializing period for the system to do the calculation. In order to minimize the error of calibration, we make the drift error an variable instead of a fixed one. Getting acceleration data of three axis and let mean of these data be the drift error, that is

$$R = (ax_1 + ax_2 + \dots + ax_n)/n \quad (14)$$

where R represents the regression line of certain axis. $ax_1 \dots ax_n$ means n axis acceleration that we used to correct acceleration when initialing the system.

Another problem we concentrate on is the drift of the acceleration when the quadrotor is moving. Kalman Filtering is one useful tool to smooth data to get better result. This

has been widely used in orbital predictions, pose estimation, etc[18][19]. Kalman combines the data from the sensor with estimated data to correct mutually. This method works well because it introduces the estimate theory into measurement and control field.

This is a method based on linear algebra and hidden Markov model (HMM), the dynamic model of the algorithm can be written as Markov chain [20]. It begins with the Linear Stochastic Difference equation :

$$X(k) = A \cdot X(k-1) + B \cdot U(k) + W(k) \quad (15)$$

and the measurement of the system:

$$Z(k) = H \cdot X(k) + V(k) \quad (16)$$

from these equations above, we can conclude the class five equations that we used.

It is easy to calculate velocity of three axis based on these acceleration data after smoothing. Noticing that velocity of some axis may have values even the acceleration of this axis equals to zero. That is because the accelerator gives a non-zero value and drops to zero in short time, according to (17) that we used to calculate velocity :

$$\int_0^t acc dt \quad (17)$$

it can not return to zero if it can't acquire the opposite value of acceleration for a period of time. Drift then begins to accumulate with time changes. Noticing that when acceleration returns to zero, the corresponding velocity can maintain a certain value for a period of time, this is different compare with flying at a certain velocity because the latter one will contains noise. To get more precise result, we amend the shift with (18), when velocity is set to zero.

$$Shift = Shift_{old} - m \times \Delta_t \times \vartheta_{off} \quad (18)$$

where ϑ_{off} represents the error that velocity hold during $m \times \Delta_t$. This is a calibration in computing displacement. Detecting m value of velocity in time, if they are equal strictly, then reset the velocity to zero. If m is small enough, it will occupy little time and not affect real-time property.

IV. EXPERIMENT

A. Crazyflie and Environment Setting

Considering the goal that we want to achieve in this project, we take the crazyflie as the testing platform, as the Fig.4 shows, this vehicle is relatively small, light and flexible, thus is useful in studying its motion. This is a quadrotor weight 19g, 9 cm $\times$ 9 cm size, and the flight time can up to 7 minutes with charging only 20 minutes[21].

Devices on board provide plenty of required data. Crazyflie carries a accelerator, magnetometer and gyroscope with high accuracy that provide motion and gesture data. The barometer makes the crazyflie get its altitude information, which is truly important for this vehicle flying at a certain height. The Cortex M3 MPU have powerful processing



Fig. 4. Crazyflie:experiment testing platform

ability. Crazyflie uses NRF2401 which works at 2.4G to communicate with ground station.

This vehicle is robust and survive in most crashes, which demonstrate it is a stable platform[22]. It cannot carry some heavy load, this may be a drawback of this vehicle. However, some light camera can be equipped on the quadrotor. In [23], a PAL video camera is taken by the vehicle to capture scene, although the image data is processed by the ground station instead of MPU onboard and retransmit the data to the quadrotor. This is a track because it can satisfied research topic with camera without considering the MPU processing capacity.

Crazyflie is an open-source platform. It has a mirror that can be installed on the virtual box, which provides the developing kit. The GUI that shows data acquired is intuitive so as to help us analyze the gesture and other parameters that interests us. On the other hand, there exits API which can help us achieve our own method. Thanks to Oliver Dunkley's work, his group developed the software as the ROS package so that we can develop the quadrotor on ROS [23]. This can ensure the compatibility and portability.

Experiments are performed in a typical environment. Indoor environment can help reduce some nature environment noise(e.g. wind). We Control the quadrotor flying along a certain line and collecting the distance data.

B. Performance

At the beginning of the system, data of IMU will be corrected by our method dynamically. As Fig.5 shows, the acceleration data contains offset, after calibrating it using mean value of acceleration data.

The other two axis have similar response like y axis. Non-fixed offset value provide wonderful little-error start for the system when it is initialized.

We compute velocity through acceleration after smoothing. As is mentioned above, the velocity can hold some wrong value because of disturbance of acceleration even it is smoothed. Correcting the velocity manually in time can avoid severe drift in distance. As Fig.6 shows, the values show excellent property of return to zero, little drift is detecting.

In order to show good property of our method ,make the quadrotor fly along a certain line drawing on the ground and collecting distance data to draw the trajectory, Fig.7 means

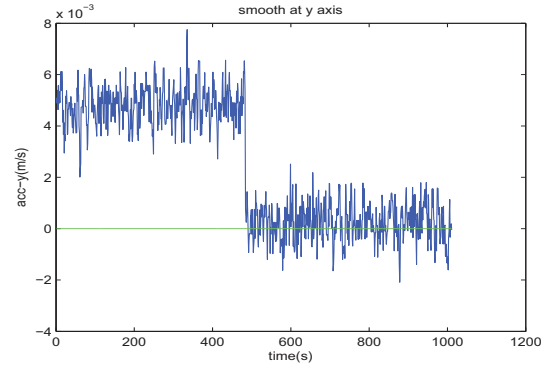


Fig. 5. Smooth acceleration data at y axis

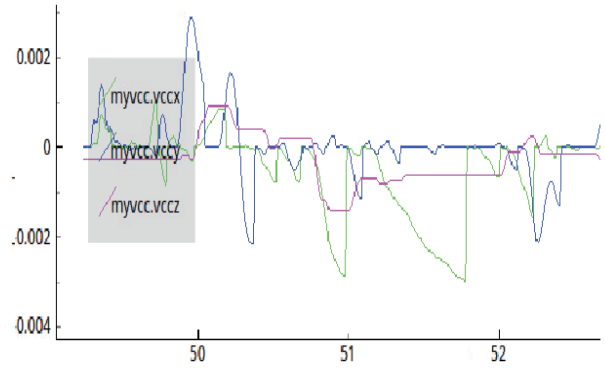


Fig. 6. Velocity detecting in real-time

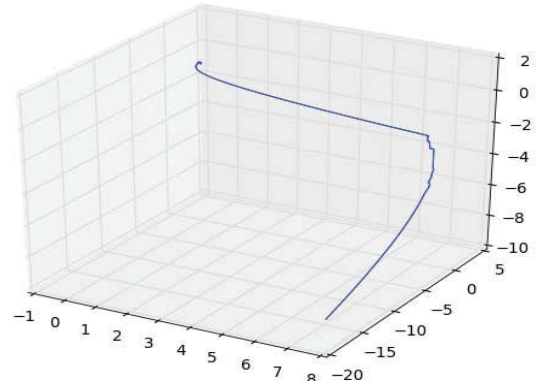


Fig. 7. Trajectory of the vehicle

the trajectory using our method is coincide with the line on the ground, which shows great property of this algorithm.

V. CONCLUSIONS AND FUTURE WORKS

This paper provides a practical method to calibrate the error in IMU so as to get data precisely. This method not only corrects the error of observed value, namely acceleration of three axis using offset-minus method and Kalman Filtering, but also detects the value of velocity and displacement and makes sure they can be amended when they show drift. This work shows higher precision which is meaningful in dead reckoning, it is a benchmark to get accurate location and track trajectory of the vehicle simply rely on IMU data. In the

long run, we can develop localization method using this data, furthermore, we can also fuse the data with other sensors, such as GPS and camera in localization.

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